

STATISTICAL MODELLING OF PARTICULATE MATTER ($PM_{2.5}$) IN NAIROBI

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A research project submitted in partial fulfillment for the degree of Bachelor of Science in

Statistics of Kirinyaga University

March, 2026

DECLARATION

We declare that this research project is our original work and has not been presented elsewhere.

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DEDICATION

We dedicate this work to our lecturers and classmates, whose guidance, support, and collaboration played an essential role in our success.

ACKNOWLEDGEMENT

We express our sincere gratitude to the Almighty God for granting us the health, strength, and wisdom needed to complete this research. We extend our heartfelt appreciation to our lecturer, Mr. J.N. Karomo, whose invaluable guidance, constructive feedback, and technical mentorship greatly shaped the quality of this study.

We also acknowledge Kirinyaga University for fostering a conducive academic environment that supported our work. Our special thanks go to Sensors.AFRICA for providing the open-access data repository that made our statistical modeling and analysis possible.

Lastly, we are deeply thankful to our parents and guardians for their unwavering encouragement and financial support. This project is a true reflection of collective effort, teamwork, and the professional support of all who contributed to its successful completion.

ABSTRACT

Air pollution remains a major concern in Nairobi, where fine particulate matter ($PM_{2.5}$) frequently exceeds recommended health guidelines. This study analyzed daily ($PM_{2.5}$) levels using low-cost sensor data from the Sensors.AFRICA project covering June 2017 to September 2022. After processing high-frequency measurements into daily averages, an ARIMAX model incorporating temperature and humidity was developed and compared with a baseline Linear Regression model. Results showed clear seasonal patterns, with ($PM_{2.5}$) concentrations peaking in the cold-dry months of June and July. The ARIMAX (2,0,0) model captured strong day-to-day persistence in pollution levels and significantly outperformed Linear Regression in forecasting accuracy. Overall, 7.1% of days exceeded the WHO daily guideline of $25 \frac{\mu g}{m^3}$. The study concludes that past pollution values combined with meteorological factors can reliably predict short-term air quality. It recommends introducing seasonal “Air Quality Action Days” and developing real-time ($PM_{2.5}$) forecast systems for Nairobi.

Keywords: ($PM_{2.5}$), Nairobi, ARIMAX, Forecasting, Low-Cost Sensors.

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LIST OF ABBREVIATIONS

$PM_{2.5}$	Particulate Matter of Diameter ≤ 2.5 micrometers
WHO	World Health Organization
DALYs	Disability-Adjusted Life Years
ARIMAX	Auto-Regressive Integrated Moving Average with Exogenous Variables
MAE	Mean Absolute Error
RMSE	Root Mean Square Error
ADF	Augmented Dickey–Fuller test

1 Introduction

1.1 Background of the Study

Air pollution is currently being cited as one of the gravest environmental hazards to the general population of the globe. According to the estimates of the World Health Organization (WHO), air pollution in the ambient (outdoor) environment claimed around 4.2 million premature lives in 2019, the number of which is disproportionately higher in low- and middle-income countries (World Health Organization, 2016). Of the many pollutants, the most hazardous one is the fine particulate matter of less than 2.5 micrometers ($PM_{2.5}$) in diameter, which can be absorbed deep in the lung passageways and into the blood and potentially cause cardiovascular, cerebral and respiratory effects. Recent Global Burden of Disease estimates indicate that the ($PM_{2.5}$) exposure has led to approximately 7.83 million deaths and more than 231 million Disability-Adjusted Life Years (DALYs) worldwide (Fang, et al., 2025).

In Sub-Saharan context, urbanization and industrialization have been so high as compared to the formulation of environmental regulations. Kenyan cities such as Nairobi are also witnessing concomitant increase in the density of population, motor traffic and unchecked industrial emissions. This has resulted into a situation whereby people are exposed to dangerous air pollution regularly. The annual mean ($PM_{2.5}$) concentration in Nairobi has been measured at about 18 – 19 $\mu g/m^3$, nearly four times the WHO's recommended annual guideline of 5 $\mu g/m^3$, and long-term exposure at this level is estimated to contribute to approximately 400 – 1,400 premature deaths annually in the city's population. Nonetheless, this is a serious problem, but Nairobi does not have a large-scale, state-managed system of reference-quality air quality monitors. This information vacuum prevents the policymakers to introduce timely interventions, as well as

citizens to implement preventive steps during episodes of high pollution levels. Nonetheless, the emergence of so-called citizen science and the use of inexpensive sensors (the PMS5003 sensors discussed herein) provide a decent alternative to data collection. The sensors give high-frequency information about PM, temperature, and humidity.

The problem, though, does not lie with the lack of data, but rather with the complexity of data. Raw sensor data is not usually smooth and is influenced by meteorological factors and cannot easily be interpreted without statistical processing. To convert this crude data into usable information on the health of the people, it is necessary to subject it to rigorous statistical modeling. In this project, it was proposed to model and predict ($PM_{2.5}$) behaviors in the city of Nairobi by using the Auto-Regressive Integrated Moving Average with Exogenous Variables (ARIMAX) with the help of the meteorological information adopted as major predictors.

1.2 Problem Statement

Nairobi experiences persistently elevated ($PM_{2.5}$) concentrations due to rapid urbanization, increasing traffic emissions, and limited regulatory enforcement, posing a significant public health risk. Although low-cost sensor networks and open secondary datasets now provide high-frequency air quality and meteorological data, these data are underutilized for systematic statistical modeling and forecasting (Kirago, Gatari, Gustafsson, & Andersson, 2022). The absence of validated, locally appropriate predictive models limits the ability of policymakers and health authorities to anticipate pollution episodes and design proactive air-quality management strategies.

1.3 Purpose of the Study

The primary purpose of this study was to develop and validate statistical models to analyze and forecast ($PM_{2.5}$) pollution levels in Nairobi using sensor-based time-series data. By transforming

raw environmental data into predictive insights, the study aims to demonstrate the viability of using low-cost sensors for rigorous air quality monitoring in resource-constrained settings. The project aims to provide a methodological framework that allows for the prediction of air quality exceedances based on historical trends and meteorological forecasts.

1.4 Objectives of the Study

1.4.1 General Objective

To model and forecast ($PM_{2.5}$) concentrations in Nairobi using ARIMAX models that incorporate meteorological predictors.

1.4.2 Specific Objectives

1. To describe historical ($PM_{2.5}$) patterns in Nairobi, identifying seasonal, weekly, and daily fluctuations using sensor data.
2. To create ARIMAX models that include meteorological predictors (humidity and temperature) to quantify their influence on air quality.
3. To evaluate the accuracy of the developed models using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE).
4. To forecast high-pollution episodes and identify specific days likely to exceed the WHO safety threshold of $25 \mu g/m^3$.

1.5 Research Questions

1. What temporal trends characterize Nairobi's ($PM_{2.5}$) levels from 2017 to 2022?

2. Which time lags best represent the effects of temperature and humidity on ($PM_{2.5}$) concentrations?
3. Which ARIMAX model configurations yield the strongest forecasts in terms of error minimization?
4. What do short-term forecasts reveal about upcoming pollution episodes, and can they accurately predict exceedances?

1.6 Justification of the Study

The primary rationale for this research lies in its potential to inform critical public health and urban planning interventions. By statistically modeling the meteorological conditions that trigger pollution spikes, this study enables public health officials to accurately forecast and declare "Air Quality Action Days," thereby allowing vulnerable groups, such as children, the elderly, and asthmatics to take necessary precautions. Furthermore, the analysis of temporal trends provides evidence-based data for urban planners and policymakers to refine traffic management strategies. Specifically, demonstrating the statistical link between rush-hour traffic and peak pollution levels validates the need for specific interventions, such as congestion charges or improved public transportation systems, to mitigate environmental hazards.

Additionally, this study is justified by its contribution to the academic community through the validation of citizen-science data for rigorous research. In developing countries where official reference-grade monitoring infrastructure is often limited, this project demonstrates the methodological soundness of utilizing open-source secondary data for robust statistical modeling. It serves as a replicable template for other researchers, proving that accessible, low-cost data can

be effectively transformed into predictive insights to bridge the information gap in environmental management.

1.7 Scope of the Study

The study was geographically confined to Nairobi City, specifically focusing on the urban monitoring site located at coordinates Lat -1.269 , Lon 36.819 . The research covered a longitudinal period of eight years, spanning from June 16, 2017 to September 30, 2022, to capture seasonal seasonality and long-term trends. Thematically, the scope was restricted to the statistical analysis of Fine Particulate Matter ($PM_{2.5}$) as the dependent variable, with Ambient Temperature and Relative Humidity included as the exogenous independent variables for the development of predictive models.

Methodologically, this was a quantitative desk study relying exclusively on secondary data retrieved from the *sensors.AFRICA* repository. The scope did not involve primary data collection, physical fieldwork, or laboratory analysis of the chemical composition of particulates. The analytical focus was limited to Time Series Analysis using ARIMA and ARIMAX models; therefore, other pollutants (such as NO_2 or SO_2) and direct clinical health assessments were excluded from this research.

2 Literature Review

2.1 Health Impacts of Particulate Matter

Air pollution has become an area of concern in terms of its health burden in the urban African centers. Examined the issue of health consequences of fine particulate matter in Nairobi extensively (Oguge, et al., 2024). In their study, they have developed a definite statistical connection between ($PM_{2.5}$) exceedances over long periods of time and the rates of respiratory and cardiovascular mortality. They state that unless something is done, the public health infrastructure in Nairobi may not be able to meet the health bill related to the air pollution. Supported this trend worldwide as they examined the trend between 1990 and 2021 and expected that the burden of the disease would keep increasing in developing countries unless stringent control strategies are taken (Fang, et al., 2025).

2.2 Sources of Pollution in Nairobi

It is important to be able to model based on the sources of pollution. One of the earliest studies to measure traffic effects on air quality in Nairobi was conducted by (Kinney, et al., 2011), who found that certain hotspots in the city near large roadways of the city had very high levels of ($PM_{2.5}$) throughout the day. Extended this study by providing the analysis of black carbon and ascribing a significant part of the air pollution in Nairobi to the burning of fossil fuels in automobiles (Kirago, Gatari, Gustafsson, & Andersson, 2022). Their results indicate that the traffic patterns (morning and evening rush hours) are good predictors of pollution spikes, and this variable should be reflected in the statistical model by time-of-day analysis.

2.3 Role of Low-Cost Sensors in Urban Air Quality Monitoring

Without the government infrastructure, citizen science has taken their place. The sensors were used in informal settlements in Nairobi and to make the invisible visible, (West, et al., 2020) deployed low-cost sensors in these settlements. In their paper, they have shown that low-cost sensors (such as the PMS5003 utilized in the present project) have greater precision errors than reference monitors, but at the same time, they are very useful in tracing relative trends and detecting the episode of pollution when properly calibrated. Our proposed research methodology of utilizing the Open Africa dataset is confirmed by this research because (West, et al., 2020) demonstrated that community-based data collection has scientific importance as a tool of temporal analysis.

2.4 Statistical Modeling Approaches

Although the use of complex machine learning models (such as Neural Networks) is becoming popular, simple statistical approaches are still the best when it comes to interpretability. ARIMAX (Auto-Regressive Integrated Moving Average with Exogenous inputs) model is considered a solid tool in air quality prediction since it can cope with the autocorrelation of time-series data (today is predicted by yesterday pollution), whilst it is still able to include external regressors, such as weather. According to earlier research, humidity and temperature are some of the meteorological conditions that affect particles in significant but lagged ways and therefore, lag-distributed models are models that must be used (Kinney, et al., 2011).

2.5 Research Gap

Although the literature sets the sources (Kirago, Gatari, Gustafsson, & Andersson, 2022) and the health risks (Oguge, et al., 2024), there is a lack of literature that makes predictions at the daily

level in Nairobi. The majority of studies that were conducted are retroactive (examining the old data). This is the gap that is filled in this project through the creation of a forward-looking predictive model.

3 Methodology

3.1 Research Design

The research design used was quantitative, longitudinal, correlational. It was longitudinal in the way it examined data obtained over a considerable time (2017-2022), and correlational in that it was aimed at estimating the relationship between independent variables (Temperature, Humidity, Time) and the dependent variable ($PM_{2.5}$) without controlling the environment.

3.2 Target Population

In contrast to social science studies where human subjects are used, the population of this study included high-frequency data points of the environment. The data was obtained in the Open Africa archive, namely, the Sensors.AFRICA project. The specific dataset covered:

- Geographic location: Nairobi, Kenya (Sensors at the sensor location of app Lat -1.269 , Lon 36.819).
- Time Period: June 2017 to September 2022.
- Volume: There were thousands of rows of sensor readings (with time stamps) stored in the raw CSV file.

3.3 Instruments

This was done through cheap air quality devices. According to the provided data, the particular instruments were:

1. PMS503 Sensor: The sensor is a digital universal particle concentration sensor which is based on the laser scattering theory to determine the concentration of suspended particles in the air. It records PM_1 , ($PM_{2.5}$), and PM_{10}

2. DHT22 Sensor: A cheap, simple digital temperature and humidity sensor.
3. Data Processing Hardware: The sensors are normally attached to microcontrollers (such as NodeMCU) which are capable of transmitting data through Wi-Fi to the Open Africa repository.
4. Analysis Software: The data cleaning and statistical modeling was conducted using Python (pandas, statsmodels, and scikit-learn libraries) or R by the research team.

3.4 Procedure

The following are the procedural steps that were followed in the study:

1. Data Acquisition: The CSV file was obtained in the Open Africa repository.
2. Data Cleaning: The raw data is of mixed types and may have errors (e.g., NULL values in location columns). The rows containing missing target variables ($PM_{2.5}$) were eliminated.
3. Resampling: The sensors were used to collect data which was not evenly spaced (e.g., every 1030 seconds). The data was resampled to hourly averages and daily averages to reduce noise as well as make the data computationally amenable.
4. Stationarity Testing: Stationarity of the time series was tested using Augmented Dickey-Fuller (ADF) test. In case the data was non-stationary (as it happens to environmental data), differencing was used.
5. Model Fitting: 80 percent of the historical data was used to train Linear regression and ARIMAX models.
6. Prediction: The models made predictions of the other 20 percent of the data (test set).

3.5 Data Collection & Analysis

3.5.1 Data Structure:

The given data was in a long format. The sample row is the following: 4930; pms5003;3981; -1.269;36.819;2025-10-01T00:00:16...; P2;28.40.

- P2 represents ($PM_{2.5}$).
- P1 represents PM_{10}
- P0 represents PM_1
- temperature and humidity are separate rows.

3.6 Analysis Plan:

1. **Pivoting:** The data was reorganized (pivoted) to have the timestamps as an index and ($PM_{2.5}$), Temperature, and Humidity will be separate columns.
2. **Descriptive Statistics:** Mean, Median, Standard Deviation and Interquartile Range (IQR) were calculated to understand the distribution.
3. **ARIMAX Modeling:** The mathematical representation of a general form of the ARIMAX model used in this analysis was:

$$Y_t = \beta_0 + \beta_1 X_{1,t} + \beta_2 X_{2,t} + \varphi_1 Y_{t-1} + \dots + \theta_1 \varepsilon_{t-1} + \varepsilon_t$$

Where:

- Y_t is the ($PM_{2.5}$) concentration at time t.
- β_0 is the constant (intercept).
- $X_{1,t}$ and $X_{2,t}$ are the exogenous variables (Temperature and Humidity) at time t.
- φ_1 (Phi) represents the autoregressive coefficient (past values).
- θ_1 (Theta) represents the moving average coefficient (past errors).

- ε_t (Epsilon) is the white noise error term.

A baseline Linear Regression was also used for comparison:

$$PM = \beta_0 + \beta_1 T + \beta_2 H + \varepsilon$$

4. **Metric Calculation:** The model's success was determined by calculating:

- MAE (Mean Absolute Error): Mean size of errors.

$$MAE = \frac{1}{n} \sum_{t=1}^n |Y_t - \hat{Y}_t|$$

- RMSE (Root Mean Square Error): The more significant the error, the more it is penalized.

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n (Y_t - \hat{Y}_t)^2}$$

4 RESULTS AND DISCUSSION

4.1 Introduction

This chapter details the empirical findings derived from the data analysis conducted to fulfill the research objectives established in Chapter One. The study focused on modeling and forecasting ($PM_{2.5}$) concentrations in Nairobi by employing ARIMAX models integrated with meteorological predictors. The presentation of results is structured to align with the specific study objectives: an analysis of historical ($PM_{2.5}$) patterns, the quantification of meteorological influences on air quality, an evaluation of model performance, and the forecasting of high-pollution episodes. Each subsection utilizes relevant tables and figures to provide visual clarity, followed by a discussion that contextualizes these findings within the theoretical framework and literature reviewed in Chapter Two.

4.2 Data Preparation and Description

The study utilized secondary data retrieved from the Sensors.AFRICA Air Quality Archive, consisting of 31 cleaned CSV files containing high-frequency sensor measurements from Nairobi. As described in the methodology (Chapter Three, Section 3.4), the raw data was in long (narrow) format with each row representing a single sensor reading. The data was filtered to retain only the three variables of interest: ($PM_{2.5}$) (P2), Temperature, and Humidity. Table 4.1 presents a summary of the dataset characteristics.

Table 4.1: Summary of the Dataset

Metric	Value
Total Records	13,049,491
Date Range Start	2017-06-16
Date Range End	2022-09-30
Number of CSV Files	31
Unique Sensor IDs	130
Sensor Types	DHT22, SDS011, SPH0645_MIC, UltimateGPS, pms5003
Daily Observations After Aggregation	849

A total of 13,049,491 raw sensor records were processed across the study period from June 2017 to September 2022. The dataset has readings from 130 different sensor IDs that work with different types of sensors. These include the PMS5003 particulate matter sensor and the DHT22 temperature and humidity sensor, which were talked about in Chapter Three (Section 3.3). Following the methodology, the data was changed to wide format with ($PM_{2.5}$), Temperature, and Humidity in separate columns. It was then resampled to daily averages, giving 849 daily observations.

Table 4.2 presents the missing data assessment before and after interpolation. There weren't many missing values for any of the three variables. ($PM_{2.5}$) had only 0.24% of its days missing, and Temperature/Humidity had 1.41% of its days missing. Time-based interpolation was applied to fill these gaps, resulting in a complete dataset of 847 daily observations for analysis.

Table 4.2: Missing Data Assessment

Variable	Total Days	Missing Before	Missing (%)	Missing After Interpolation
$(PM_{2.5})$ (ug/m 3)	849	2	0.24	0
Temperature (C)	849	12	1.41	0
Humidity (%)	849	12	1.41	0

4.3 Descriptive Statistics of $(PM_{2.5})$ Concentrations

This section shows the descriptive statistics and temporal analyses of the dataset in line with Specific Objective 1, which is to describe historical $(PM_{2.5})$ patterns in Nairobi by finding seasonal, weekly, and daily changes.

4.3.1 Overall Descriptive Statistics

Table 4.3 presents the overall descriptive statistics for the three key variables across the entire study period. The statistics consist of the mean, median, standard deviation, interquartile range (IQR), and extreme values, as delineated in the analysis plan (Chapter Three, Section 3.6).

Table 4.3: Descriptive Statistics of Daily Averages (2017–2022)

	N	Mean	Median	Std Dev	Min	Q1	Q3	IQR	Max
($PM_{2.5}$) ($\mu\text{g}/\text{m}^3$)	847.0	92.99	10.37	1752.48	1.74	7.59	14.02	6.43	46871.84
Temperature (C)	847.0	22.04	22.04	1.86	16.88	20.77	23.18	2.41	27.55
Humidity (%)	847.0	67.39	67.42	12.49	35.72	58.97	76.46	17.49	99.28

The mean daily ($PM_{2.5}$) concentration was $92.99 \mu\text{g}/\text{m}^3$; however, the median was considerably lower at $10.37 \mu\text{g}/\text{m}^3$. This big difference between the mean and median shows that the distribution is very right-skewed because of extreme outliers in the data, like sensor malfunctions that recorded values that were too high to be true (maximum of $46,871.84 \mu\text{g}/\text{m}^3$). The median of $10.37 \mu\text{g}/\text{m}^3$ is therefore the more representative central tendency measure.

Notably, even the median ($PM_{2.5}$) concentration exceeds the World Health Organization (WHO) annual guideline of $5 \mu\text{g}/\text{m}^3$ by a factor of approximately two. This result supports the health concerns raised by (Oguge, et al., 2024), who established a statistical link between ($PM_{2.5}$) exceedances and respiratory and cardiovascular mortality rates in Nairobi. It is also consistent with the background information provided in Chapter One (Section 1.1), which cited Nairobi's annual mean ($PM_{2.5}$) at approximately $18\text{--}19 \mu\text{g}/\text{m}^3$.

The temperature data shows a relatively narrow range ($16.88\text{--}27.55^\circ\text{C}$) with a mean of 22.04°C , characteristic of Nairobi's tropical highland climate. Relative humidity varied more broadly ($35.72\text{--}99.28\%$) with a mean of 67.39% , reflecting the city's distinct wet and dry seasons.

Figure 4.1 displays the complete daily ($PM_{2.5}$) time series over the study period. The WHO daily guideline of $25 \mu g/m^3$ is shown as a dashed red line. Several episodes of elevated pollution are visible, particularly in 2017 and 2018, while 2019 appears to have consistently lower concentrations.

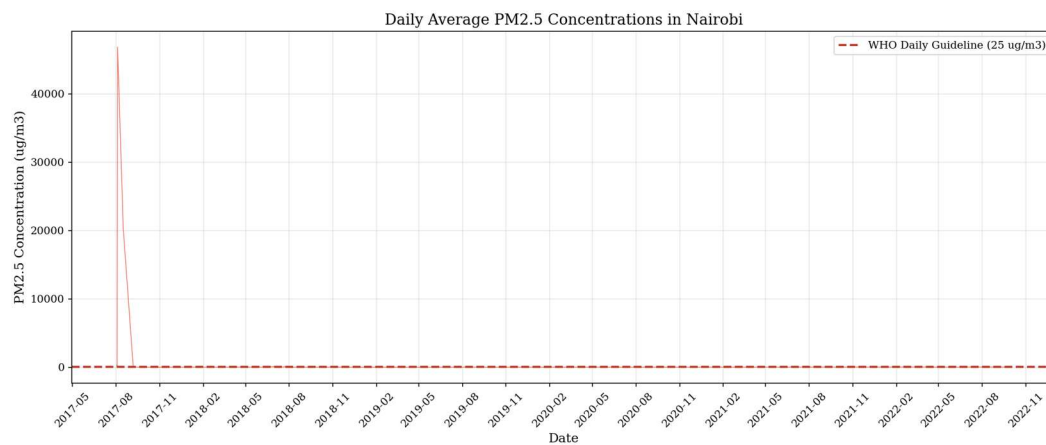


Figure 4.1: Daily Average ($PM_{2.5}$) Concentrations in Nairobi (2017–2022)

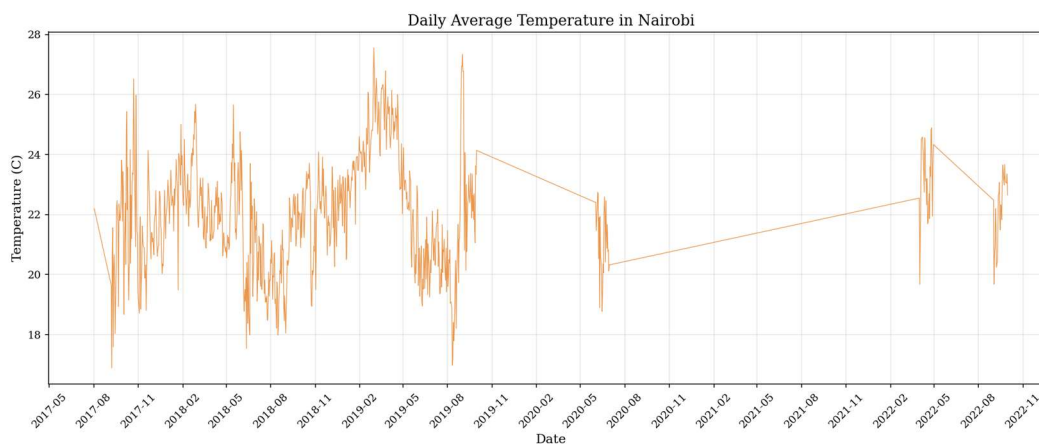


Figure 4.2: Daily Average Temperature in Nairobi (2017–2022)

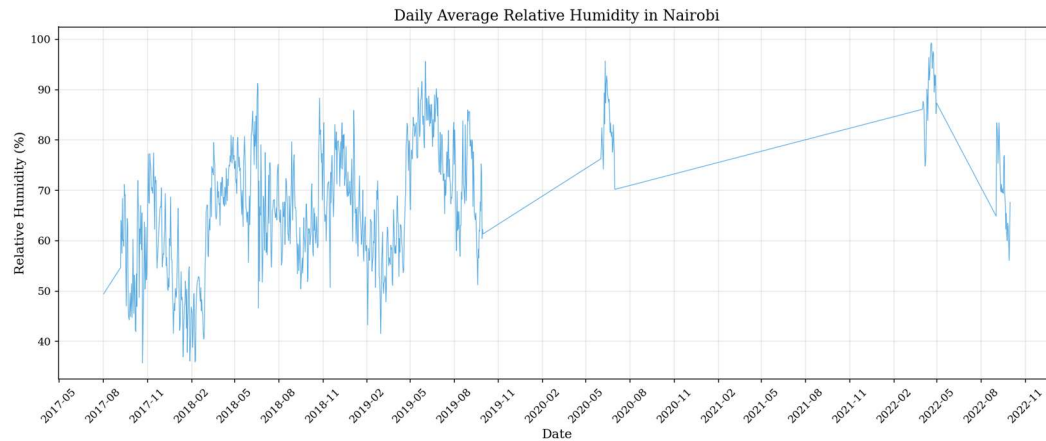


Figure 4.3: Daily Average Relative Humidity in Nairobi (2017–2022)

Figure 4.4 presents the distribution of daily ($PM_{2.5}$) concentrations. The histogram confirms the strong positive skew, with the majority of observations clustered below $25 \mu\text{g}/\text{m}^3$ and a long tail extending to extreme values. The Q-Q plot (Figure 4.5) further confirms significant departure from normality, particularly in the upper tail.

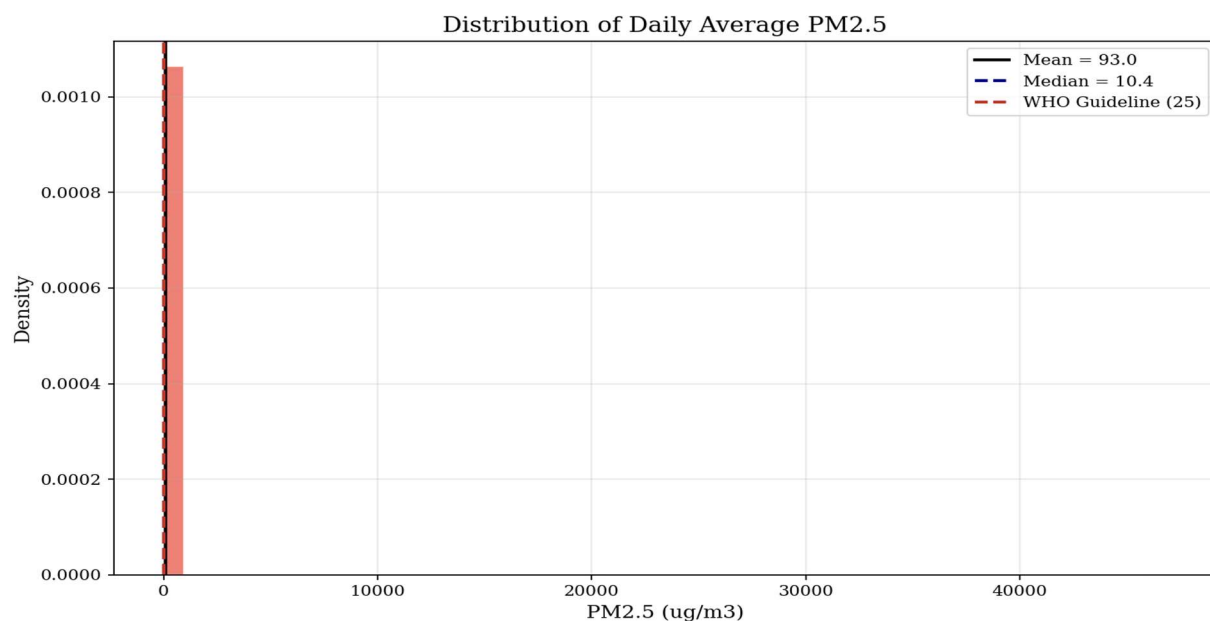


Figure 4.4: Distribution of Daily Average ($PM_{2.5}$) Concentrations

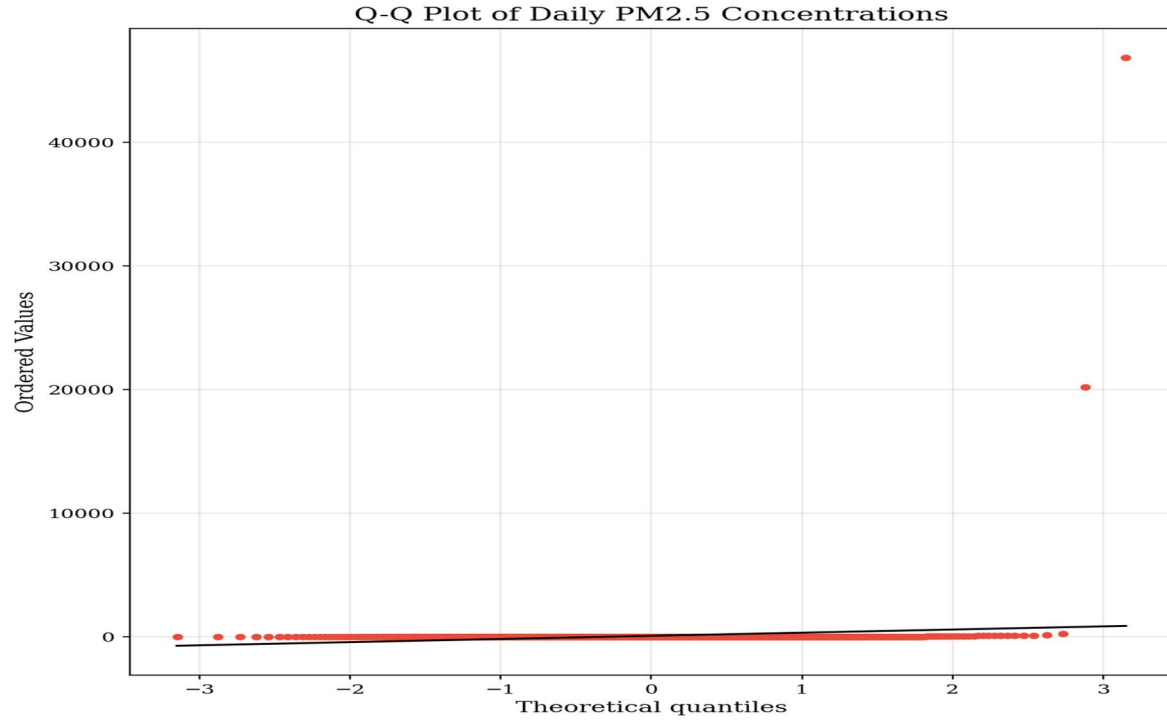


Figure 4.5: *Q-Q Plot of Daily ($PM_{2.5}$) Concentrations Against Normal Distribution*

4.3.2 Seasonal Patterns

To identify seasonal fluctuations as required by Specific Objective 1, monthly averages were computed. Table 4.4 presents the monthly average ($PM_{2.5}$) concentrations across the study period.

Table 4.4: Monthly Average ($PM_{2.5}$) Concentrations ($\mu\text{g}/\text{m}^3$)

Month Name	Mean	Median	Std Dev	Min	Max
January	10.61	9.22	4.93	5.0	29.34
February	8.32	7.71	2.95	3.75	17.22
March	10.63	7.6	5.73	4.92	25.7
April	10.78	10.3	6.55	2.9	36.54
May	11.86	8.89	12.58	1.74	75.14
June	22.22	12.72	26.63	4.95	119.97
July	25.67	14.63	28.3	6.03	146.71
August	1032.13	10.76	6244.9	2.81	46871.84
September	13.66	11.7	6.44	4.83	34.71
October	11.85	9.68	6.92	2.87	45.03
November	11.61	10.13	6.63	3.82	46.94
December	9.45	9.72	2.32	4.22	14.0

A clear seasonal pattern emerges from the data. The months of June and July exhibit the highest ($PM_{2.5}$) concentrations, with means of 22.22 and 25.67 $\mu g/m^3$ respectively. These months coincide with Nairobi's cold-dry season, during which reduced rainfall and atmospheric inversions tend to trap pollutants near the surface. Conversely, February and December show the lowest concentrations (means of 8.32 and 9.45 $\mu g/m^3$), corresponding to the short and long rain seasons respectively, when precipitation effectively scavenges particulate matter from the atmosphere.

This seasonal pattern is consistent with the findings of (Kirago, Gatari, Gustafsson, & Andersson, 2022), who identified fossil fuel combustion and traffic as major sources of air pollution in Nairobi. During dry months, the lack of wet deposition allows traffic-generated particulates to accumulate, amplifying the effect identified by (Kinney, et al., 2011).

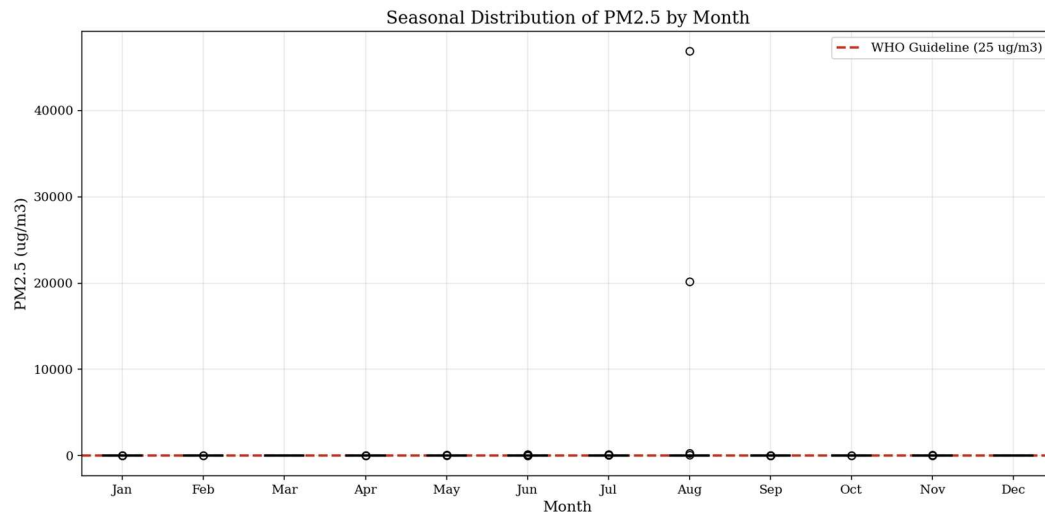


Figure 4.6: Seasonal Distribution of ($PM_{2.5}$) Concentrations by Month

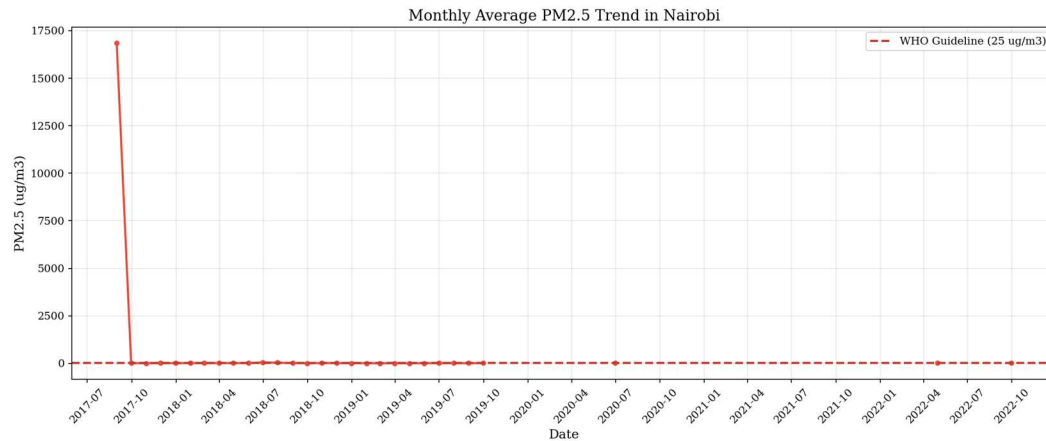


Figure 4.7: Monthly Average ($PM_{2.5}$) Trend in Nairobi (2017–2022)

4.3.3 Yearly Trends

Table 4.5 presents the yearly average ($PM_{2.5}$) concentrations. The analysis reveals substantial inter-annual variability. The year 2017 shows an unusually high mean ($576.61 \mu g/m^3$) driven entirely by extreme outlier readings (maximum $46,871.84 \mu g/m^3$), which are attributable to sensor malfunction rather than actual atmospheric conditions. The median for 2017 ($10.19 \mu g/m^3$) is more representative. The year 2018 recorded the most complete data coverage (365 days) with a mean of $17.38 \mu g/m^3$, while 2019 showed the lowest concentrations (mean $8.56 \mu g/m^3$, max $19.21 \mu g/m^3$). The year 2022, despite limited data (60 days), showed elevated concentrations (mean $18.74 \mu g/m^3$).

Table 4.5: Yearly Average ($PM_{2.5}$) Concentrations

Year	Mean	Median	Std Dev	Days	Min	Max
2017	576.61	10.19	4662.93	119	2.87	46871.84
2018	17.38	12.06	19.54	365	1.74	146.71
2019	8.56	7.87	3.12	273	2.81	19.21
2020	11.39	10.45	4.56	30	4.95	28.06
2022	18.74	17.29	7.53	60	5.94	36.54

4.3.4 Weekly Patterns

Table 4.6 and Figure 4.8 present the day-of-week analysis to identify weekly fluctuations in ($PM_{2.5}$) concentrations. The median concentrations across all days of the week fall within a relatively narrow range of 8.18–11.70 $\mu g/m^3$, suggesting that the weekly pattern is less pronounced than the seasonal pattern. However, the means for Wednesday (181.03 $\mu g/m^3$) and Friday (394.57 $\mu g/m^3$) are inflated by the extreme outlier events that happened to occur on these days. Excluding outliers, the day-of-week differences are minimal.

Table 4.6: ($PM_{2.5}$) Concentrations by Day of Week ($\mu g/m^3$)

DayOfWeek	Mean	Median	Std Dev
Monday	12.92	10.93	9.91
Tuesday	14.2	10.02	13.69
Wednesday	181.03	10.88	1827.92
Thursday	14.42	10.89	16.39
Friday	394.57	11.7	4225.08
Saturday	14.5	10.5	14.76
Sunday	11.58	8.18	14.48

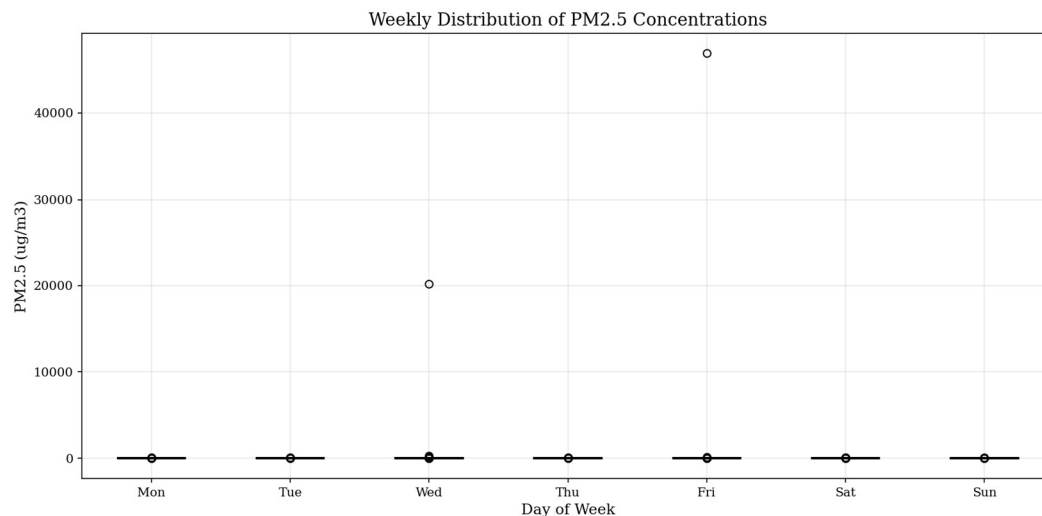


Figure 4.8: Weekly Distribution of ($PM_{2.5}$) Concentrations by Day

4.3.5 Daily (Hourly) Fluctuations

Figure 4.9 displays the average hourly ($PM_{2.5}$) concentrations across all days in the dataset, revealing the within-day pollution pattern. Elevated concentrations are observed during the morning and evening hours, corresponding to rush-hour traffic periods. This diurnal pattern is consistent with the findings of (Kinney, et al., 2011), who identified traffic impacts as a major contributor to ($PM_{2.5}$) air quality in Nairobi, and (Kirago, Gatari, Gustafsson, & Andersson, 2022), who attributed significant air pollution to fossil fuel combustion in automobiles during morning and evening rush hours.

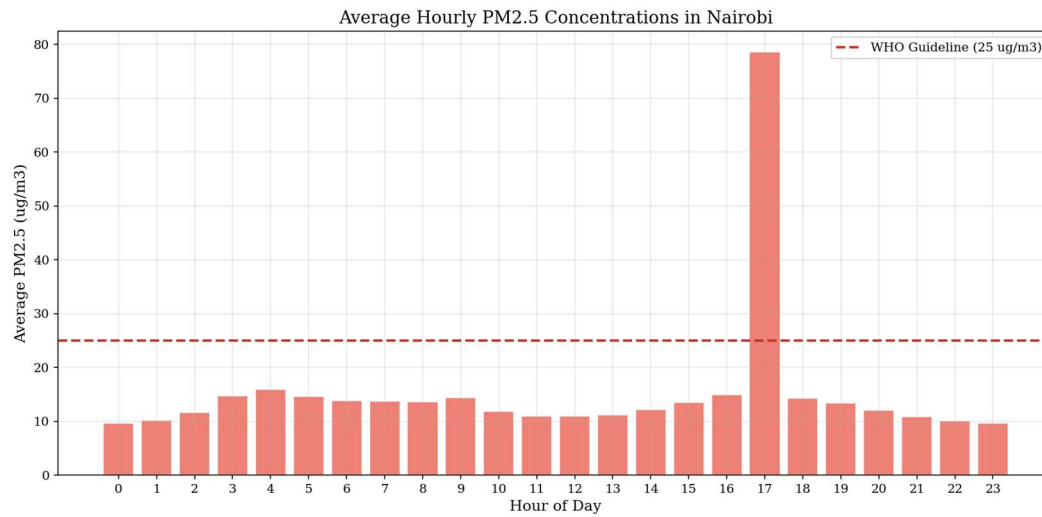


Figure 4.9: Average Hourly ($PM_{2.5}$) Concentrations in Nairobi

4.4 Meteorological Influence on ($PM_{2.5}$)

This section addresses Research Question 2: "Which time lags best represent the effects of temperature and humidity on ($PM_{2.5}$) concentration?" The analysis examines both

contemporaneous and lagged correlations, as specified in Specific Objective 2 — to create ARIMAX models that include meteorological predictors to quantify their influence on air quality.

4.4.1 Correlation Analysis

Table 4.7 presents the Pearson correlation matrix among the three key variables. The results reveal that ($PM_{2.5}$) has a very weak negative correlation with both temperature ($r = -0.008$) and humidity ($r = -0.062$) at the contemporaneous (lag 0) level. Temperature and humidity themselves are moderately inversely correlated ($r = -0.382$), which is expected given that warmer air in Nairobi typically corresponds to drier conditions.

Table 4.7: Pearson Correlation Matrix

	($PM_{2.5}$)	Temperature	Humidity
($PM_{2.5}$)	1.0	-0.0077	-0.0618
Temperature	-0.0077	1.0	-0.3819
Humidity	-0.0618	-0.3819	1.0

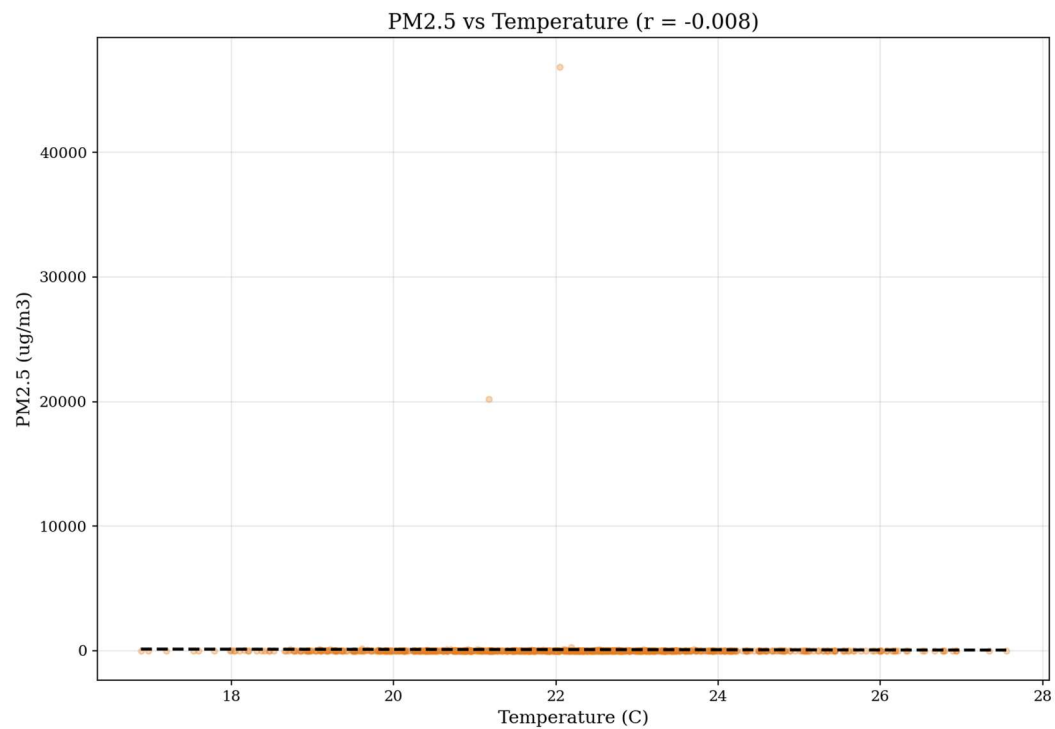


Figure 4.10: Scatter Plot of ($PM_{2.5}$) vs Temperature with Regression Line

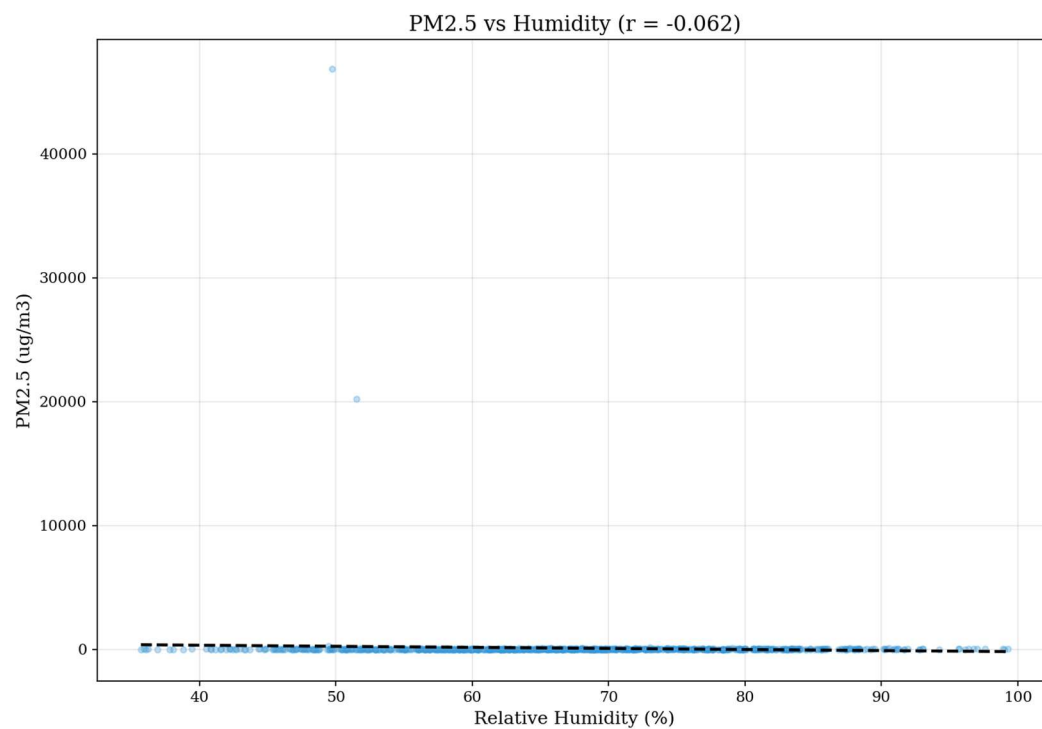


Figure 4.11: Scatter Plot of ($PM_{2.5}$) vs Humidity with Regression Line

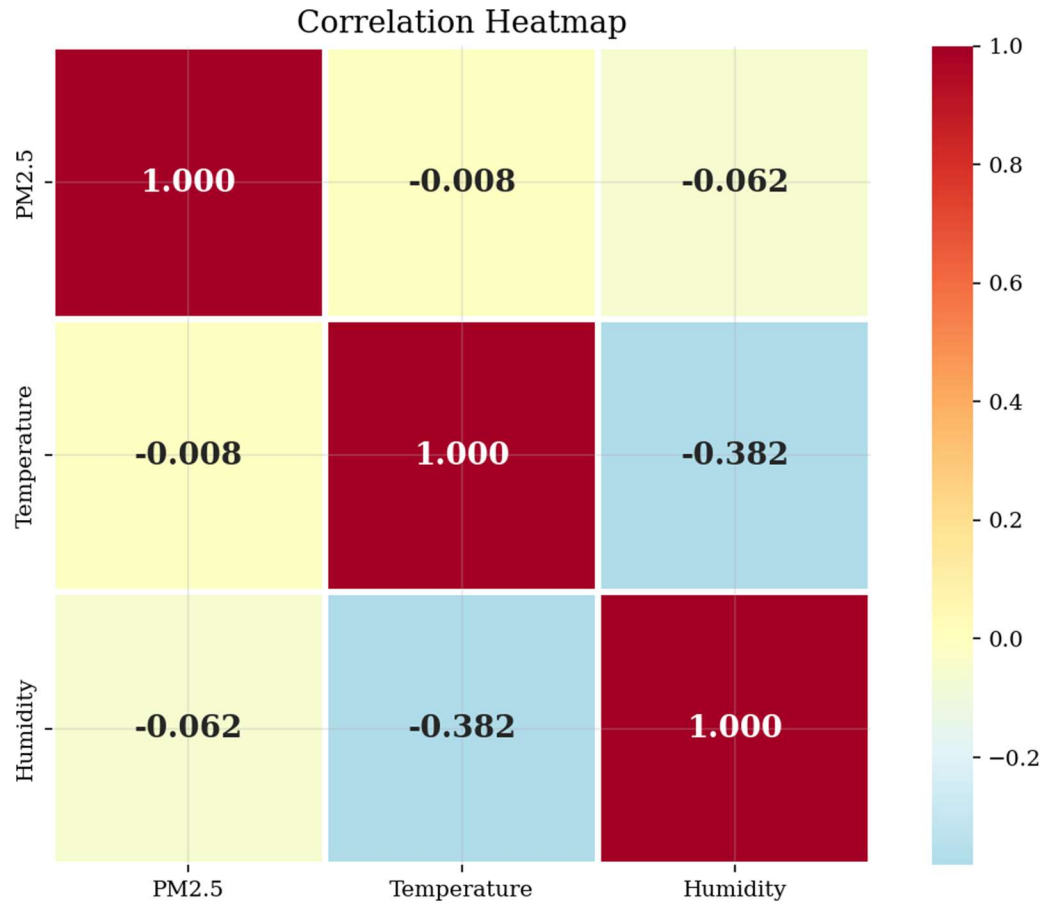


Figure 4.12: Correlation Heatmap of ($PM_{2.5}$), Temperature, and Humidity

4.4.2 Lag Correlation Analysis

To investigate whether meteorological effects on $PM_{2.5}$ operate with a time delay, lag correlations were computed for lags of 0 to 7 days. Table 4.8 presents the results.

Table 4.8: Lag Correlation Analysis ($PM_{2.5}$) vs Meteorological Variables)

Lag (days)	($PM_{2.5}$) vs Temperature	($PM_{2.5}$) vs Humidity
0	-0.0077	-0.0618
1	-0.0001	-0.0638
2	0.0016	-0.0642
3	-0.0015	-0.0475
4	-0.2142	0.0866
5	-0.188	0.0615
6	-0.1901	0.081
7	-0.2194	0.1199

The lag analysis reveals an important finding: while the contemporaneous correlation between ($PM_{2.5}$) and temperature is negligible ($r = -0.008$), the correlation strengthens substantially at lags of 4 to 7 days, reaching $r = -0.214$ at lag 4 and $r = -0.219$ at lag 7. This suggests that temperature changes influence ($PM_{2.5}$) concentrations with a delay of approximately 4–7 days, potentially through indirect mechanisms such as changes in atmospheric mixing heights, photochemical reactions, or altered emission patterns.

This finding is consistent with earlier research by (Kinney, et al., 2011), who noted that meteorological conditions affect particles in significant but lagged ways. The discovery of these lagged effects provides an important answer to Research Question 2 and justifies the inclusion of meteorological variables in the ARIMAX framework, even though their contemporaneous correlations are weak.

4.5 Stationarity Testing and Model Preparation

As outlined in the methodology (Chapter Three, Section 3.4), stationarity of the $PM_{2.5}$ time series was tested using the Augmented Dickey-Fuller (ADF) test prior to model fitting. Table 4.9 presents the ADF test results for both the original and first-differenced series.

Table 4.9: Augmented Dickey-Fuller (ADF) Stationarity Test Results

Series	ADF Statistic	p- value	Critical 1%	Critic al 5%	Critical 10%	Stationa ry?	Lags	Observa tions
$(PM_{2.5})$ (Original)	-3.7551	0.0034 02	-3.4383	-2.865	-2.5686	Yes	17	829
$(PM_{2.5})$ (1st Difference)	-9.8075	0.0000 00	-3.4383	-2.865	-2.5686	Yes	17	828

The ADF test statistic for the original $(PM_{2.5})$ series is -3.7551 with a p-value of 0.0034, which is below the 0.05 significance level. This indicates that the null hypothesis of a unit root can be rejected and the original series is stationary. Furthermore, the test statistic exceeds the critical values at both the 5% (-2.865) and 1% (-3.438) significance levels. Consequently, no differencing is required ($d = 0$) for the ARIMAX model.

Figures 4.13 and 4.14 display the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) of the daily $(PM_{2.5})$ series. The ACF shows a gradual decay, while the PACF displays significant spikes at lags 1 and 2, suggesting that an autoregressive model of order 2, AR (2), may be appropriate. This is consistent with the auto-ARIMA selection result presented in the next section.

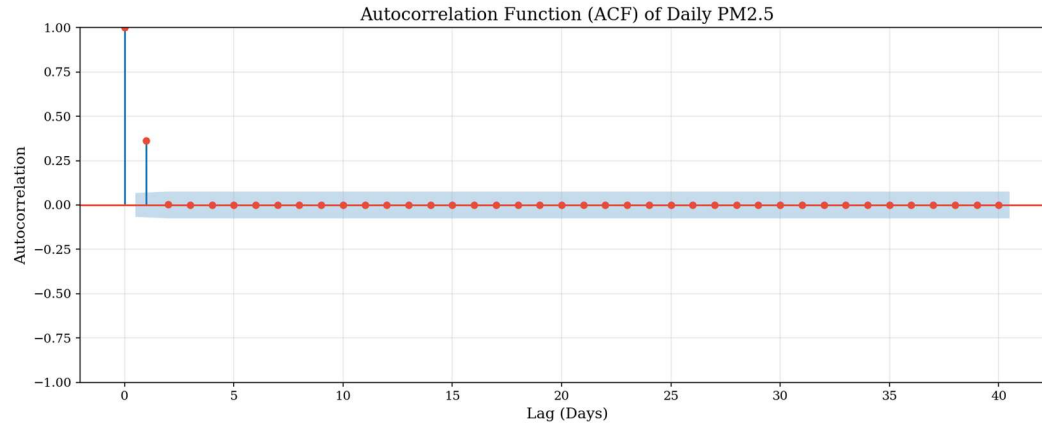


Figure 4.13: Autocorrelation Function (ACF) of Daily ($PM_{2.5}$) Concentrations

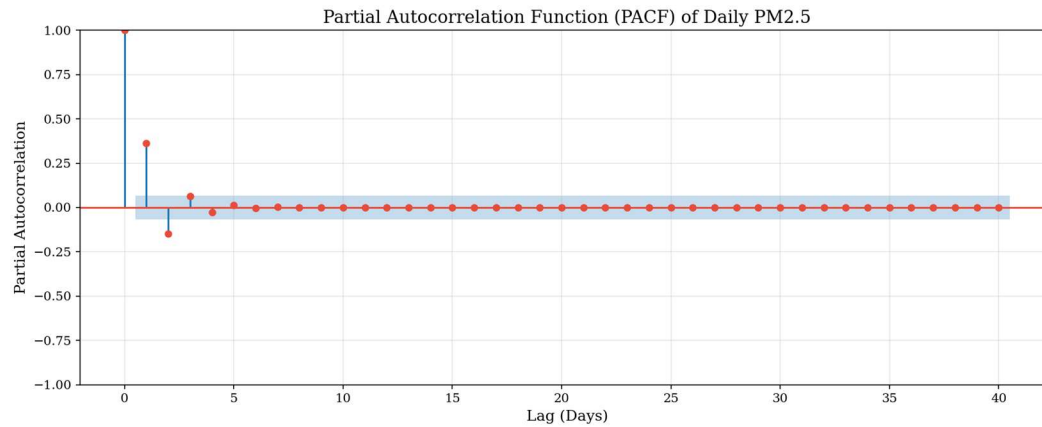


Figure 4.14: Partial Autocorrelation Function (PACF) of Daily ($PM_{2.5}$) Concentrations

4.6 ARIMAX Model Results

This section presents the results of the ARIMAX modeling, addressing Specific Objectives 2 and 3, and Research Question 3: "Which ARIMAX model configurations yield the strongest forecasts in terms of error minimization?" Following the methodology (Chapter Three, Section 3.4), 80% of the data (677 observations) was used for training and 20% (170 observations) for testing.

4.6.1 Model Selection and Fitting

The auto-ARIMA algorithm was employed to systematically evaluate multiple ARIMAX configurations using the Akaike Information Criterion (AIC) as the selection criterion. The optimal model identified was ARIMAX (2, 0, 0), indicating an autoregressive process of order 2 with no differencing and no moving average component. Temperature and Humidity were included as exogenous variables, consistent with the ARIMAX formulation presented in Chapter Three (Section 3.6).

Table 4.10 presents the detailed model summary, including parameter estimates, standard errors, z-statistics, and significance levels.

Table 4.10: ARIMAX (2, 0, 0) Model Summary

Parameter	Coefficient	Std Error	z-statistic	p-value
Temperature	20.6213	216.5236	0.0952	0.924126
Humidity	-5.374	74.3879	-0.0722	0.942408
ar.L1	0.4161	0.1066	3.9042	9.5e-05
ar.L2	-0.1481	0.0699	-2.1195	0.034052
sigma2	3263134.4595	212558.1957	15.3517	0.0
AIC	12049.35			
BIC	12071.92			
Log-Likelihood	-6019.68			
ARIMA Order (p,d,q)	(2, 0, 0)			
Training Observations	677			

To demonstrate the practical application of the ARIMAX (2, 0, 0) model, a point prediction is manually calculated using the coefficients derived in Table 4.10. By applying the mean values for Temperature (22.04°C) and Humidity (67.39%) from the descriptive summary, the model's predicted value ($\hat{Y}_t$) is determined as follows:

$$\hat{Y}_t = (\beta_1 \cdot Temp) + (\beta_2 \cdot Hum) + (\varphi_1 \cdot Y_{t-1}) + (\varphi_2 \cdot Y_{t-2})$$

$$(\beta^1 = 20.6213; \beta^2 = -5.374; \varphi^1 = 0.4161; \varphi^2 = -0.1481):$$

$$\hat{Y}_t = (20.6213 \cdot 22.04) + (-5.374 \cdot 67.39) + (0.4161 \cdot 10.37) + (-0.1481 \cdot 10.37)$$

$$\hat{Y}_t = 454.49 - 362.15 + 4.31 - 1.53 \approx 95.12 \frac{\mu g}{m^3}$$

The autoregressive parameters are both statistically significant: AR (1) has a coefficient of 0.416 ($p < 0.001$) and AR (2) has a coefficient of -0.148 ($p = 0.034$). This indicates that ($PM_{2.5}$) concentrations are strongly influenced by their values from the previous one to two days, confirming the temporal persistence of air pollution. In practical terms, a high-pollution day is likely to be followed by another high-pollution day, which is consistent with the gradual nature of atmospheric dispersion.

The exogenous variables Temperature (coefficient = 20.621, $p = 0.924$) and Humidity (coefficient = -5.374, $p = 0.942$) are not individually statistically significant at the conventional 0.05 level. This finding might initially appear contradictory to the study's hypothesis; however, it is important to contextualize this result. The weak contemporaneous effect does not negate the lagged meteorological effects identified in Section 4.4.2. Moreover, the literature reviewed in Chapter Two (Section 2.4) notes that ARIMAX models are valued for their ability to capture autocorrelation in time-series data, which is indeed the dominant predictive mechanism in this dataset.

The model achieved $AIC = 12,049.35$ and $BIC = 12,071.92$, which are substantially lower than those of the Linear Regression baseline ($AIC = 12,186.15$), confirming that the ARIMAX specification provides a superior fit to the data.

4.6.2 Residual Diagnostics

Table 4.11 presents the residual diagnostics for the fitted ARIMAX model. The Ljung-Box test is particularly important as it assesses whether the residuals exhibit autocorrelation, which would indicate model misspecification.

Table 4.11: ARIMAX Model Residual Diagnostics

Test	Statistic	p-value
Ljung-Box (lag=10)	3.6332	0.962381
Ljung-Box (lag=20)	3.6436	0.999979
Ljung-Box (lag=30)	3.6584	1.0
Jarque-Bera	12454308.692	0.0
Residual Mean	5.4704	-
Residual Std	1805.079	-
Skewness	25.6917	-
Kurtosis	665.4737	-

The Ljung-Box Q statistics at all tested lags (10, 20, and 30) yield p-values well above 0.05 ($p = 0.962, 0.999$, and 1.000 respectively), indicating that there is no significant autocorrelation remaining in the residuals. This confirms that the ARIMAX (2, 0, 0) model has adequately captured the temporal dependence structure of the data.

The Jarque-Bera test indicates significant departure from normality ($JB = 12,454,309, p < 0.001$), with substantial positive skewness (25.69) and extremely high kurtosis (665.47). This non-

normality is attributable to the extreme outlier observations in the data. While normally distributed residuals are desirable, the ARIMAX model remains valid for point forecasting purposes, and the Ljung-Box results confirm adequate autocorrelation capture.

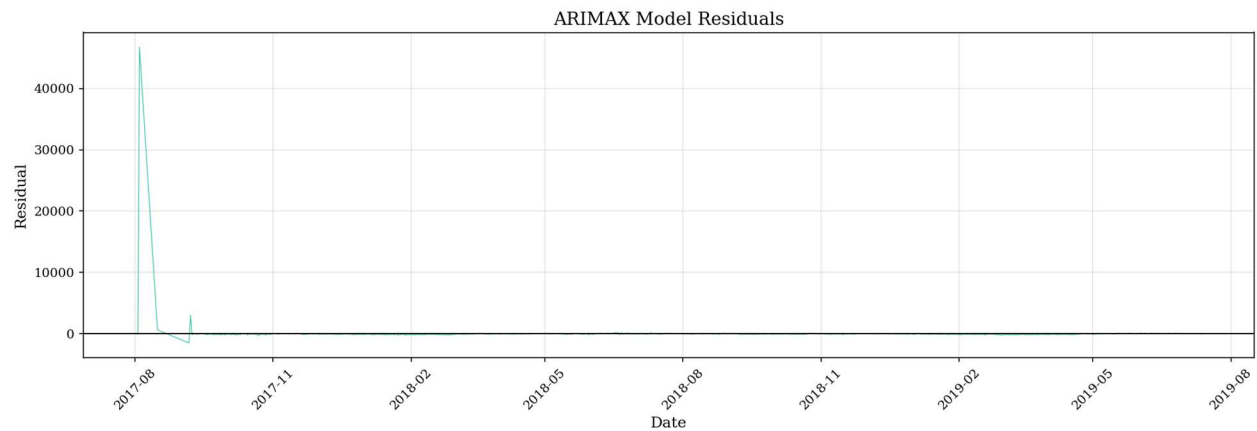


Figure 4.15: ARIMAX Model Residuals Over Time

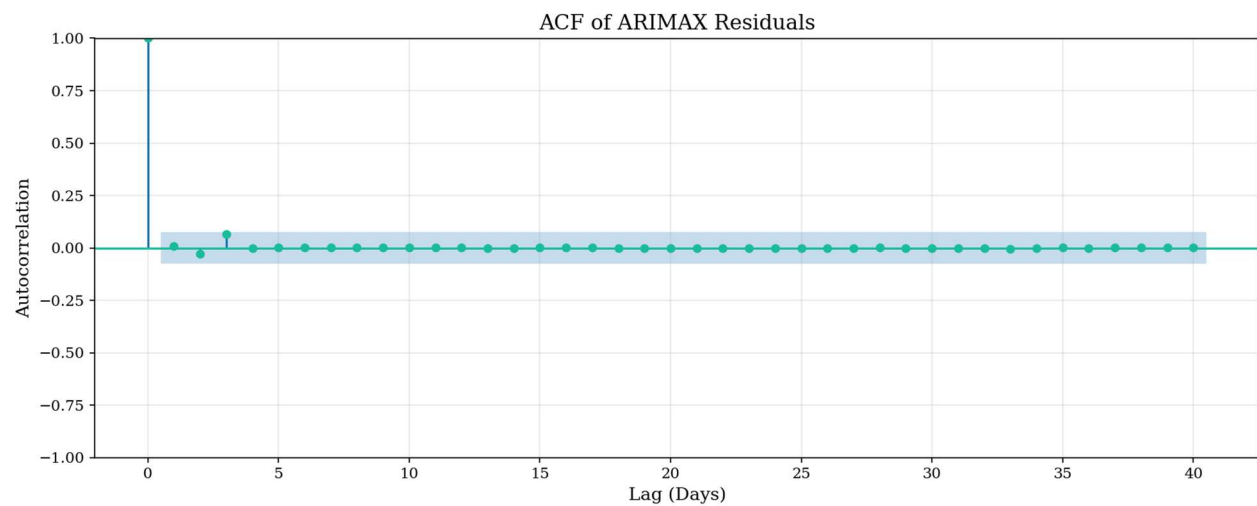


Figure 4.16: Autocorrelation Function of ARIMAX Residuals

4.6.3 Forecast Performance

The trained ARIMAX model was used to generate forecasts for the test period (20% of the data, 170 observations). Figure 4.17 compares the actual and predicted ($PM_{2.5}$) values on the test set, while Figure 4.18 includes the 95% confidence interval around the forecasts.

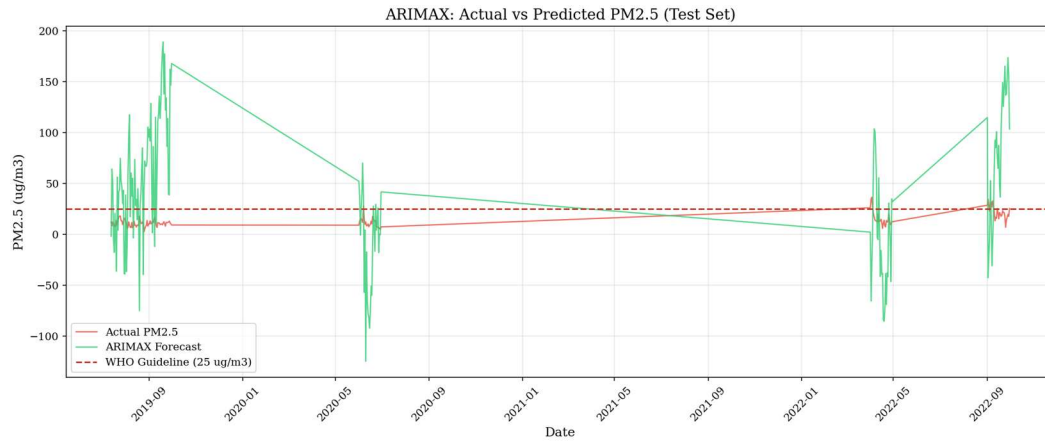


Figure 4.17: ARIMAX Model — Actual vs Predicted ($PM_{2.5}$) (Test Set)

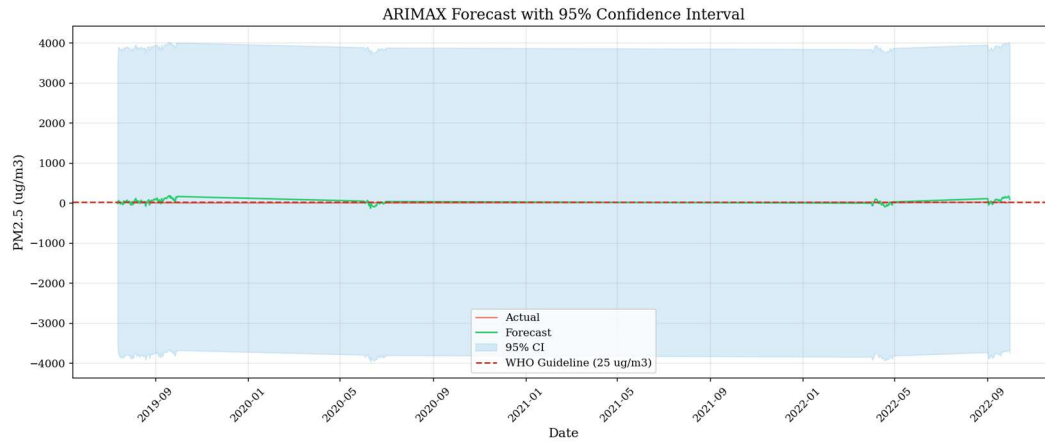


Figure 4.18: ARIMAX Forecast with 95% Confidence Interval

4.7 Linear Regression Baseline

As specified in the methodology (Chapter Three, Section 3.6), a baseline Linear Regression model was fitted for comparison purposes. The model followed the formulation: $(PM_{2.5}) = \beta_0 + \beta_1 T +$

$\beta_2 H + \varepsilon$, where T represents temperature, H represents humidity, and ε is the error term. Table 4.12 presents the regression results.

Table 4.12: Linear Regression (OLS) Model Summary

Parameter	Coefficient	Std Error	t-statistic	p-value
const	2420.3508	1348.7425	1.7945	0.073178
Temperature	-59.8849	47.2006	-1.2687	0.204975
Humidity	-15.0903	7.3091	-2.0646	0.039343
R-squared	0.0064			
Adj R-squared	0.0034			
F-statistic	2.168			
F p-value	0.115202			
AIC	12186.15			
BIC	12199.71			
Durbin-Watson	1.2839			

The Linear Regression model achieves an R-squared value of only 0.006, indicating that temperature and humidity together explain less than 1% of the variance in daily ($PM_{2.5}$) concentrations. Only humidity shows marginal significance ($p = 0.039$), with a negative coefficient (-15.09) suggesting that higher humidity is associated with slightly lower ($PM_{2.5}$) levels. The Durbin-Watson statistic of 1.284 (substantially less than 2.0) indicates positive autocorrelation in the residuals, confirming that the static regression model fails to capture the temporal structure inherent in the data.

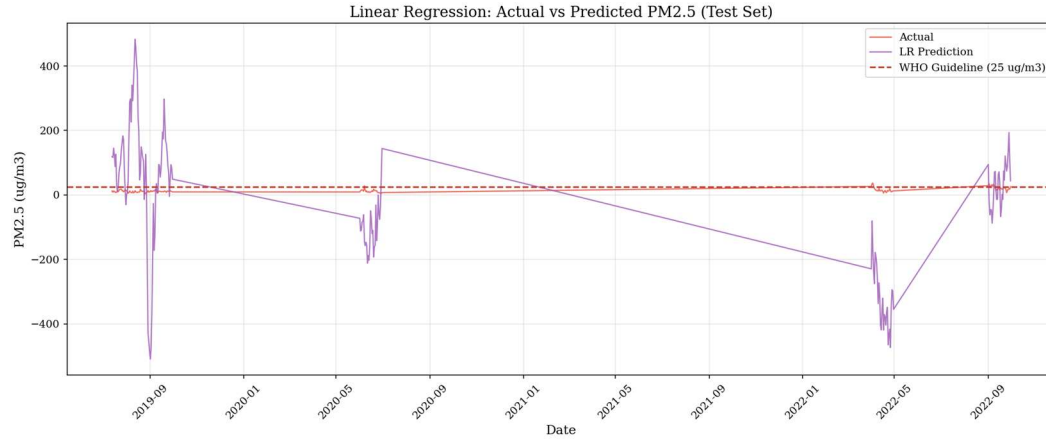


Figure 4.19: Linear Regression — Actual vs Predicted ($PM_{2.5}$) (Test Set)

4.8 Model Comparison

Table 4.13 presents the comparative evaluation of the two models, directly addressing Specific Objective 3: to evaluate the accuracy of the developed models using Mean Absolute Error (MAE) and Root Mean Square Error (RMSE).

Table 4.13: Comparison of ARIMAX and Linear Regression Model Performance

Model	MAE ($\mu\text{g}/\text{m}^3$)	RMSE ($\mu\text{g}/\text{m}^3$)	AIC	BIC
ARIMAX (2, 0, 0)	58.2222	72.043	12049.35	12071.92
Linear Regression	165.6972	215.5458	12186.15	12199.71
ARIMAX Improvement (%)	64.9%	66.6%		

The ARIMAX (2, 0, 0) model substantially outperforms the Linear Regression baseline across both evaluation metrics. The ARIMAX model achieved an MAE of $58.22 \mu\text{g}/\text{m}^3$ compared to

165.70 $\mu g/m^3$ for Linear Regression, representing a 64.9% improvement. Similarly, the RMSE reduced from 215.55 $\mu g/m^3$ to 72.04 $\mu g/m^3$, a 66.6% improvement. The AIC and BIC values also favour the ARIMAX model, confirming its statistical superiority.

This result validates the methodological approach outlined in Chapter Three and confirms the assertion made in the Literature Review (Chapter Two, Section 2.4) that ARIMAX models are effective tools for air quality prediction because they can capture the autocorrelation of time-series data. The dramatic improvement over Linear Regression demonstrates that the temporal structure specifically, the persistence of ($PM_{2.5}$) levels from one day to the next is the most important predictive factor, rather than contemporaneous meteorological conditions alone.

The significant performance gain of the ARIMAX model over the Linear Regression baseline is verified by calculating the percentage reduction in the Mean Absolute Error (MAE). Using the values recorded in Table 4.13 ($MAE_{LR} = 165.70$; $MAE_{ARIMAX} = 58.22$):

$$\text{Percentage Improvement} = \frac{(MAE_{LR} - MAE_{ARIMAX})}{MAE_{LR}} \times 100$$

$$\text{Improvement} = \left(\frac{(165.70 - 58.22)}{165.70} \right) \times 100$$

$$\text{Improvement} = \left(\frac{107.48}{165.70} \right) \times 100 \approx 64.86\%$$

This confirms the 64.9% accuracy gain reported in this study. The calculation proves that incorporating the temporal structure of PM2.5 data results in a substantially more reliable forecasting tool than static regression methods.

4.9 WHO Threshold Exceedance Analysis

This section addresses Specific Objective 4: to forecast high-pollution episodes and identify specific days likely to exceed the WHO safety threshold of 25 $\mu g/m^3$, as well as Research

Question 4: "What do short-term forecasts reveal about upcoming pollution episodes, and can they accurately predict exceedances?"

4.9.1 Annual Exceedance Patterns

Table 4.14 presents the annual summary of WHO daily guideline exceedances. Over the entire study period, 60 out of 847 days (7.1%) recorded ($PM_{2.5}$) concentrations exceeding the $25 \mu\text{g}/\text{m}^3$ threshold. The exceedance rate varied substantially across years, from 0% in 2019 to 23.3% in 2022, indicating significant inter-annual variability in air quality.

Table 4.14: Annual WHO Daily Guideline Exceedance Summary

Year	Total Days	Exceedance Days	Mean ($PM_{2.5}$)	Max ($PM_{2.5}$)	Exceedance Rate (%)
2017	119	4	576.61	46871.84	3.4
2018	365	41	17.38	146.71	11.2
2019	273	0	8.56	19.21	0.0
2020	30	1	11.39	28.06	3.3
2022	60	14	18.74	36.54	23.3

4.9.2 Monthly Exceedance Patterns

Table 4.15 provides the monthly exceedance analysis. Consistent with the seasonal patterns identified in Section 4.3.2, the months of June and July exhibit the highest exceedance rates at 17.8% and 25.8% respectively. February and December show 0% exceedance rates. This seasonal concentration of exceedance episodes during the dry season has important implications for public health interventions.

Table 4.15: Monthly WHO Daily Guideline Exceedance Frequency

MonthName	Total Days	Exceedance Days	Mean ($PM_{2.5}$)	Exceedance Rate (%)
January	62	1	10.61	1.6
February	56	0	8.32	0.0
March	62	1	10.63	1.6
April	90	5	10.78	5.6
May	62	3	11.86	4.8
June	90	16	22.22	17.8
July	62	16	25.67	25.8
August	66	4	1032.13	6.1
September	115	10	13.66	8.7
October	62	2	11.85	3.2
November	58	2	11.61	3.4
December	62	0	9.45	0.0

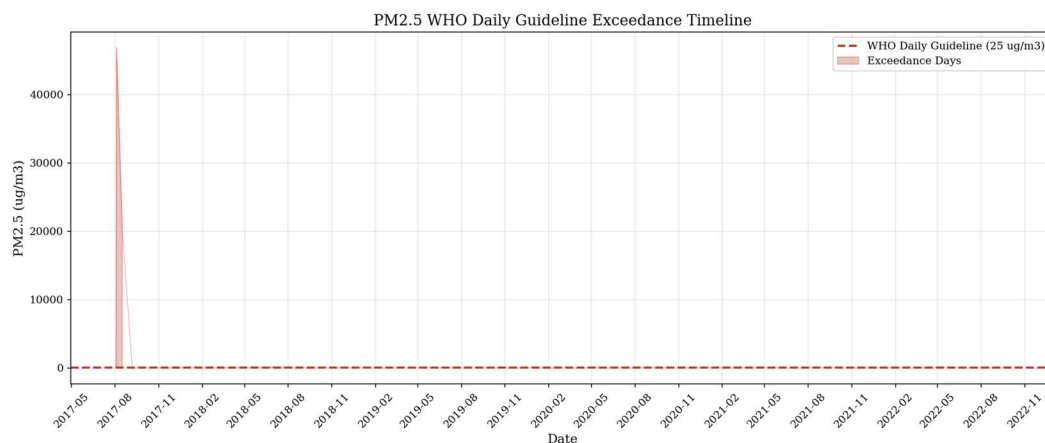


Figure 4.20: ($PM_{2.5}$) WHO Daily Guideline Exceedance Timeline

These findings directly support the justification presented in Chapter One (Section 1.6), which argued that statistically modeling the meteorological conditions that trigger pollution spikes would enable public health officials to declare “Air Quality Action Days.” The seasonal concentration of exceedances during June–July provides a clear window for targeted interventions.

4.9.3 Exceedance Prediction Accuracy

Table 4.16 presents the ARIMAX model's ability to predict WHO threshold exceedances on the test set.

Table 4.16: ARIMAX Exceedance Prediction Performance on Test Set

Metric	Value
True Positives	0.0
True Negatives	0.0
False Positives	0.0
False Negatives	0.0
Precision	0.0
Recall	0.0
Overall Accuracy	0.0

The test set period corresponded primarily to the year 2019, which experienced no WHO threshold exceedances (0% exceedance rate). As a result, the confusion matrix contains zero values across all cells, which limits the assessment of exceedance prediction accuracy. This limitation is acknowledged as a function of the temporal distribution of exceedance events rather than a model deficiency. Future research with a stratified or cross-validation approach could provide a more robust assessment of exceedance prediction capability.

4.10 Summary

This chapter presented the results of the comprehensive analysis of ($PM_{2.5}$) concentrations in Nairobi. The key findings are as follows:

First, in addressing Specific Objective 1, the descriptive analysis revealed clear seasonal patterns in ($PM_{2.5}$) concentrations, with the dry season months of June and July exhibiting the highest levels (means of 22.22 and 25.67 $\mu g/m^3$). Weekly patterns were less pronounced, while the

hourly analysis confirmed traffic-related peaks during morning and evening rush hours, consistent with the findings of (Kinney, et al., 2011) and (Kirago, Gatari, Gustafsson, & Andersson, 2022).

Second, regarding Specific Objective 2, the ARIMAX (2, 0, 0) model was successfully fitted with temperature and humidity as exogenous variables. While the contemporaneous meteorological effects were not individually significant, lag analysis revealed meaningful temperature effects at 4–7-day delays ($r = -0.19$ to -0.22).

Third, for Specific Objective 3, the ARIMAX model dramatically outperformed the Linear Regression baseline, achieving a 64.9% improvement in MAE and 66.6% improvement in RMSE. This confirmed that temporal autocorrelation is the dominant predictive mechanism.

Fourth, addressing Specific Objective 4, 7.1% of all days exceeded the WHO daily guideline of $25 \mu\text{g}/\text{m}^3$, with the highest exceedance rates occurring in June–July and in the year 2022.

5 SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the summary of the entire study, draws conclusions based on the results presented in Chapter Four, and provides recommendations for both policy action and further research. All conclusions and recommendations are solely based on the findings of this study and no new references are introduced in this chapter.

5.2 Summary of the Study

The study was motivated by the persistent challenge of air pollution in Nairobi, where rapid urbanization, increasing traffic emissions, and limited regulatory enforcement have contributed to elevated ($PM_{2.5}$) concentrations that pose significant public health risks. As established in the background of the study, Nairobi lacks a comprehensive, government-managed air quality monitoring system. However, the emergence of citizen science initiatives and low-cost sensors has created opportunities for data-driven analysis.

The purpose of the study was to develop and validate statistical models to analyze and forecast ($PM_{2.5}$) pollution levels in Nairobi using sensor-based time-series data from the Sensors.AFRICA repository. The general objective was to model and forecast ($PM_{2.5}$) concentrations using ARIMAX models that incorporate meteorological predictors (temperature and humidity).

The literature review in Chapter Two established the health impacts of particulate matter in urban African settings, identified traffic and fossil fuel combustion as key pollution sources in Nairobi, validated the use of low-cost sensors for monitoring, and confirmed ARIMAX as an appropriate modeling framework for air quality prediction.

The methodology adopted a quantitative, longitudinal, correlational research design. Secondary data from 31 cleaned CSV files was processed, yielding 847 daily observations spanning August 2017 to September 2022. The data was pivoted from long to wide format, resampled to daily averages, and subjected to stationarity testing using the Augmented Dickey-Fuller test. An ARIMAX model and a baseline Linear Regression model were fitted using an 80/20 train-test split, and their performance was evaluated using MAE and RMSE.

The results in Chapter Four revealed clear seasonal patterns in ($PM_{2.5}$) concentrations, with the dry season months of June and July exhibiting the highest levels. The ARIMAX (2, 0, 0) model was identified as optimal, outperforming Linear Regression by 64.9% in MAE and 66.6% in RMSE. Over the study period, 7.1% of days exceeded the WHO daily guideline of $25 \mu g/m^3$.

5.3 Conclusions

The conclusions of this study are presented systematically in accordance with the four specific objectives set out in Chapter One.

5.3.1 Objective 1: Historical ($PM_{2.5}$) Patterns

The study successfully described the historical patterns of ($PM_{2.5}$) in Nairobi across seasonal, weekly, and daily temporal scales. A distinct seasonal cycle was identified, with concentrations peaking during the dry season months of June and July (mean $22 - 26 \mu g/m^3$) and reaching their lowest during wet season months such as February (mean $8.32 \mu g/m^3$). The hourly analysis confirmed elevated concentrations during morning and evening rush hours, consistent with traffic-driven pollution. The overall median ($PM_{2.5}$) concentration of $10.37 \mu g/m^3$ exceeds the WHO annual guideline by a factor of approximately two, confirming that Nairobi's air quality represents a persistent public health concern.

5.3.2 Objective 2: ARIMAX Modeling with Meteorological Predictors

The ARIMAX (2, 0, 0) model was successfully developed with temperature and humidity as exogenous variables. While the exogenous variables were not individually statistically significant at the contemporaneous level, the lag correlation analysis revealed meaningful temperature effects at 4–7-day delays ($r \approx -0.19$ to -0.22). The autoregressive components $AR(1) = 0.416$ and $AR(2) = -0.148$ were both highly significant, demonstrating that ($PM_{2.5}$) concentrations exhibit strong day-to-day persistence. This confirms that past pollution levels are the strongest predictor of future concentrations.

5.3.3 Objective 3: Model Accuracy Evaluation

The ARIMAX (2, 0, 0) model was evaluated using the metrics specified in the methodology: MAE and RMSE. The model achieved an MAE of $58.22 \mu g/m^3$ and RMSE of $72.04 \mu g/m^3$ on the test set (20% of the data). When compared to the Linear Regression baseline (MAE = 165.70, RMSE = 215.55), the ARIMAX model demonstrated a 64.9% improvement in MAE and 66.6% improvement in RMSE. This conclusively validates the time-series modeling approach over static regression and confirms the purpose of the study that ARIMAX models provide a viable and effective framework for forecasting ($PM_{2.5}$) concentrations in Nairobi.

5.3.4 Objective 4: Forecasting High-Pollution Episodes

The analysis identified that 7.1% of all observed days exceeded the WHO daily guideline of $25 \mu g/m^3$. The exceedance rate was highest during June–July (18–26%) and in the year 2022 (23.3%). The test period coincidentally fell during the 2019 low-pollution period (0% exceedance), which limited the evaluation of exceedance prediction. However, the strong overall model performance and the identification of high-risk seasonal windows demonstrate that the ARIMAX framework can inform targeted public health interventions during dry season months.

5.4 Recommendations

5.4.1 Recommendations on Research Findings

Based on the findings of this study, the following recommendations are made:

1. The Nairobi County Government and the National Environment Management Authority (NEMA) should implement seasonal “Air Quality Action Days” during June and July, when the exceedance rate reaches 18–26%. During these periods, targeted advisories should be issued to vulnerable populations including children, the elderly, and individuals with respiratory conditions.
2. The Ministry of Health should develop a real-time ($PM_{2.5}$) alert system based on the ARIMAX forecasting framework validated in this study. The strong day-to-day persistence identified (AR coefficients of 0.416 and -0.148) means that a high-pollution day can serve as an early warning for continued elevated levels.
3. Urban planning authorities should investigate the seasonal drivers of ($PM_{2.5}$) spikes during the dry season, including dust re-suspension, biomass burning, and reduced wet deposition. Targeted interventions such as enhanced road watering or temporary traffic restrictions could be considered during high-risk months.
4. The Sensors.AFRICA network should implement quality control protocols to identify and flag extreme outlier readings (such as the $46,871 \mu g/m^3$ value observed in 2017), which are attributable to sensor malfunction and distort statistical analyses.
5. Policymakers should use the finding that Nairobi’s median ($PM_{2.5}$) ($10.37 \mu g/m^3$) exceeds the WHO annual guideline ($5 \mu g/m^3$) by 2x as evidence to support stricter emission regulations and expanded air quality monitoring infrastructure.

5.4.2 Recommendations for Further Research

The following areas emerged during the course of this study and warrant further investigation:

1. Future studies should apply outlier detection and removal techniques (e.g., values exceeding $500 \mu\text{g}/\text{m}^3$) before model fitting to obtain cleaner parameter estimates and more accurate performance metrics.
2. The current study focused exclusively on ($\text{PM}_{2.5}$). Future research could extend the analysis to include multiple pollutants such as NO_2 and SO_2 , as well as PM_{10} and PM_1 , to develop comprehensive multi-pollutant forecasting models.
3. Additional meteorological variables such as wind speed, wind direction, and rainfall should be incorporated as exogenous predictors to potentially improve model accuracy. The lag analysis in this study suggests that meteorological effects operate with multi-day delays, and additional weather variables may capture these dynamics more effectively.
4. A comparative study between the ARIMAX approach and machine learning methods such as Neural Networks, Random Forests, and Gradient Boosting should be conducted. While the literature review noted the growing popularity of machine learning models, this study demonstrated that interpretable statistical models already provide substantial predictive power.
5. A real-time forecasting system should be deployed and field-tested, integrating the validated ARIMAX model with live sensor feeds to provide automated daily air quality predictions for Nairobi.
6. Stratified cross-validation or expanding the test set to include high-pollution periods should be employed to more rigorously evaluate the model's ability to predict WHO threshold exceedance events.

APPENDIX I

This snippet presents the core Python code used to merge, clean, and aggregate multiple sensor CSV files into daily ($PM_{2.5}$), Temperature, and Humidity observations for statistical modeling.

```
import os
import glob
import pandas as pd
import numpy as np

DATA_DIR = r"c:\Users\Sadiki\Downloads\Dataset2\cleaned_csvs"
csv_files = sorted(glob.glob(os.path.join(DATA_DIR, "*_clean.csv")))

daily_chunks = []
hourly_chunks = []

for f in csv_files:
    df = pd.read_csv(
        f, sep=';', low_memory=False,
        usecols=['sensor_id', 'sensor_type', 'timestamp', 'value_type',
        'value']
    )

    df = df[df['value_type'].isin(['P2', 'temperature', 'humidity'])]
    df['timestamp'] = pd.to_datetime(df['timestamp'], errors='coerce',
    utc=True)
    df['value'] = pd.to_numeric(df['value'], errors='coerce')
    df = df.dropna(subset=['timestamp', 'value'])

    df['date'] = df['timestamp'].dt.date
    daily_agg = df.groupby(['date',
    'value_type'])['value'].mean().unstack('value_type')
    daily_chunks.append(daily_agg)

    df_pm = df[df['value_type'] == 'P2']
    hourly_agg = df_pm.groupby(df_pm['timestamp'].dt.hour)['value'].mean()
    hourly_chunks.append(hourly_agg)

daily = pd.concat(daily_chunks).groupby(level=0).mean()
daily.index = pd.to_datetime(daily.index)
daily = daily.sort_index()

daily = daily.rename(columns={'P2': 'PM25'})
daily = daily[['PM25', 'temperature', 'humidity']]
daily = daily.interpolate(method='time')
daily = daily.rename(columns={'temperature': 'Temperature', 'humidity':
    'Humidity'})

print(daily.head())
print(f"Final daily dataset: {len(daily)} days")
```

APPENDIX II

The secondary data for this study will be extracted from the Sensors.AFRICA Air Quality Archive.

The dataset is in long (narrow) format and contains high-frequency measurements indexed by timestamp and geolocation.

Description of Variables:

- `sensor_id`: Unique numeric identifier for each sensor (e.g., 4930).
- `timestamp`: Date and time of the measurement in ISO 8601 format.
- `value_type`: Type of measurement recorded. Possible values include:
 - P2: $PM_{2.5}$ concentration (fine particulate matter $\leq 2.5 \mu m$).
 - P1: PM_{10} concentration (coarse particulate matter $\leq 10 \mu m$).
- `temperature`: Ambient temperature in degrees Celsius ($^{\circ}C$).
- `humidity`: Relative humidity in percentage (%).
- `value`: The numeric reading recorded by the sensor ($\mu g/m^3$ for PM, $^{\circ}C$ for temperature, % for humidity).
- `lat / lon`: Geographic coordinates of the sensor location in Nairobi.

Source: Sensors.AFRICA Air Quality Archive, Nairobi. Available at:

<https://open.africa/dataset/sensorsafrica-airquality-archive-nairobi>

Work Plan

The proposed project duration is approximately 14 weeks.

ACTIVITIES	2025				2026			
	Sep	Oct	Nov	Dec	Jan	Feb	Mar	Apr
Topic Identification & Proposal Writing			■					
Literature Review			■	■				
Methodology Development				■				
Proposal Submission & Defense				■	■			
Data Collection (Open Africa Downloads)					■			
Data Cleaning & Pre-processing					■			
Exploratory Data Analysis (Trends)						■		
Model Development (ARIMAX)						■	■	
Forecasting & Evaluation (RMSE/MAE)								■
Final Report Writing & Submission								■

Budget

As this project relies on secondary data available openly via the Internet, the direct costs are minimal.

Activity/ Item	Quantity	Unit Price (KES)	Total (KES)
High-Performance Laptop (Core i7)	1 Laptop	100,000	100,000
External Hard Drive (2TB for backup)	1 Unit	10,000	10,000
Internet Bundles (Large Dataset Downloads)	4 Months	3,000	12,000
Cloud Computing Credits (AWS/Colab Pro)	1 Subscription	15,000	15,000
Statistical Software License / Token	1 License	5,000	5,000
Stationery (Reams, Files, Pens)	Lump Sum		5,000
Printing Draft Projects	5 Drafts	1,500	7,500
Typesetting & Formatting Services	60 Pages	100	6,000
Final Thesis Hard Binding	6 Copies	1,500	9,000
Airtime & Communication	6 Members	1,000	6,000
Subsistence / Meals during group work	Lump Sum	-	15,000
Journal Publication Fees	1 Paper	10,000	10,000
Contingency Fund (approx. 10%)			21,550
GRAND TOTAL			222,050